



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to AMMAN PETROLEUM

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

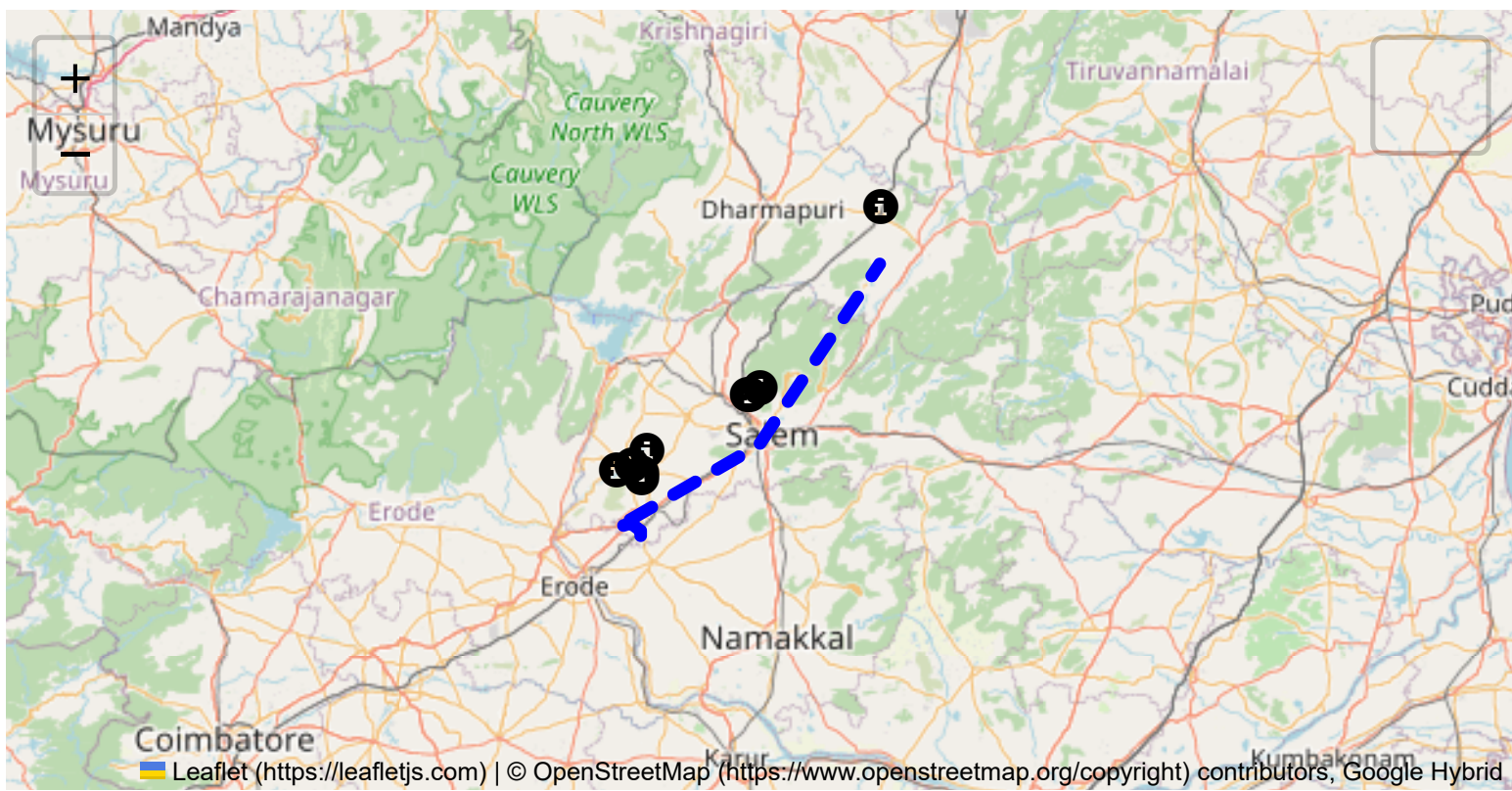
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

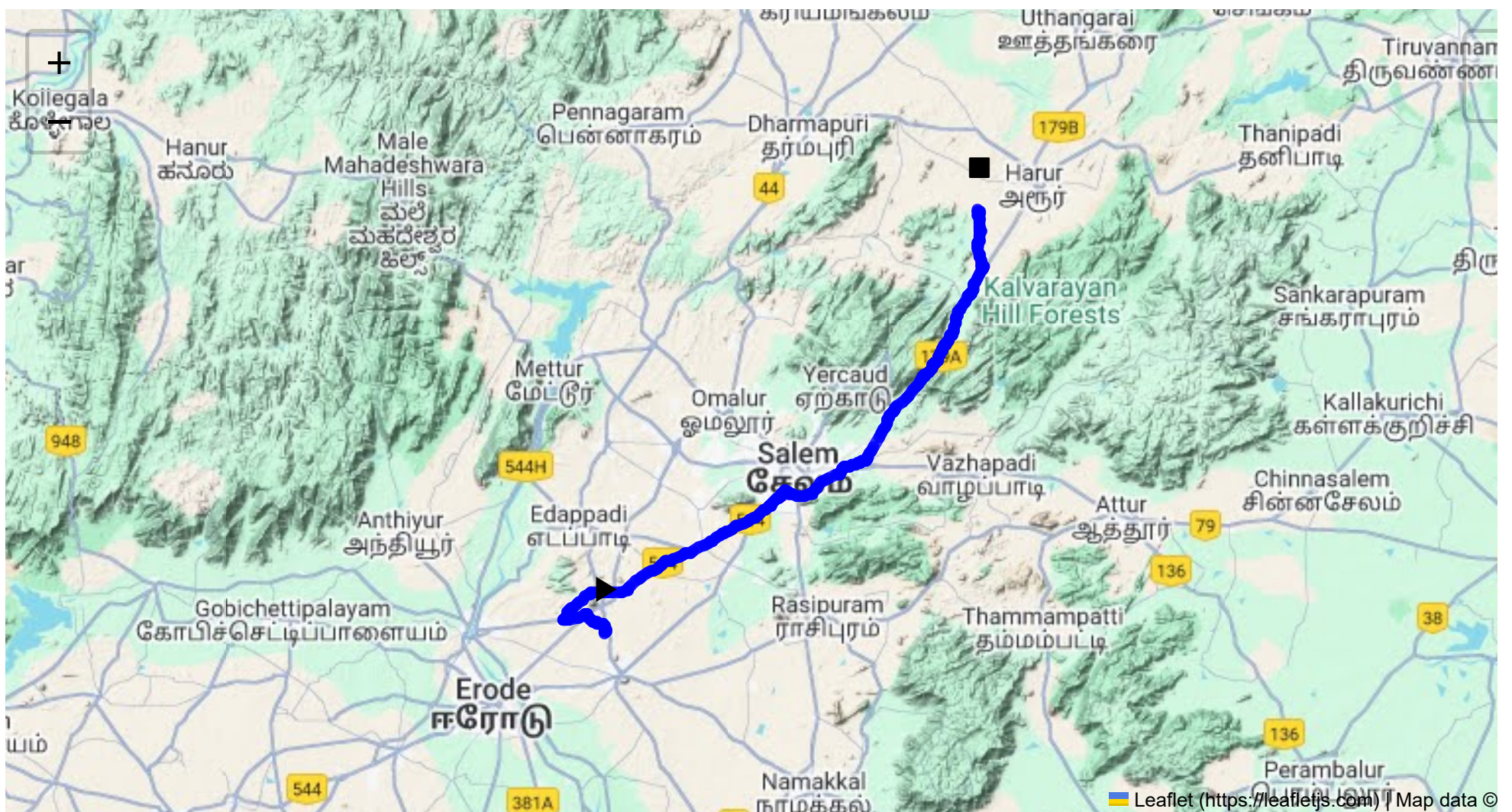
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 108.32 km**  
**Estimated Duration: 2.6 hours**  
**Adjusted Duration (Heavy Vehicle): 3.2 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.01645, 78.3948)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to Thiruvalluvarnagar, Pappireddipatti spans approximately 108.32 kilometers and passes through various regions including Sankari, Salem, and Neykkarapatti. The primary route involves NH544 and Bhavani Main Road, with sections passing through urban and semi-urban areas.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

The typical weather in Tamil Nadu includes a hot climate with temperatures ranging from 25°C to 35°C. The monsoon season brings heavy rainfall from June to September, which can lead to waterlogged roads and landslides in certain areas, particularly on highways. Drivers should be cautious of reduced visibility and slippery roads during the monsoon.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) see major congestion, particularly near urban centers like Salem.
- **Congestion-Prone Areas:** NH544, especially near Salem, and the Sankari RS - Erode Privu Road.
- **Suggestions:** Travel during non-peak hours is advisable to avoid significant delays.

## 4. Assessment of Road Quality and Infrastructure

- **NH544:** Generally good quality, but prone to bottlenecks near Salem.
- **Bhavani Main Road:** Varies in quality with some sections requiring careful navigation due to potholes.
- **Urban Areas:** Streets can be narrow with occasional encroachments in Salem and Sankari.

## 5. Suggestions for Alternative Routes for Emergencies

Alternative routes via smaller state highways parallel to NH544 can be used in case of major disruptions. Specific alternatives would require real-time traffic data for optimal selection.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations for the transport of hazardous materials, including permits for travel through certain zones and time restrictions to reduce traffic impact. Drivers must ensure all necessary paperwork is in order and adhere to prescribed routes and timings.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

This region has experienced incidents primarily due to overloading and speeding. Specific black spots, especially near Salem, have recorded accidents involving heavy vehicles. Proper load management and adherence to speed limits can mitigate risks.

## 8. Environmental Considerations and Sensitive Areas

The route traverses through agricultural regions and some semi-urban settlements. Avoiding accidental spills is crucial to protect local water bodies and farmlands. Enhanced vigilance near populated and environmentally sensitive zones is essential.

## 9. Analysis of Communication Coverage

Mobile network coverage is generally strong in urban areas but may exhibit some dead zones in rural stretches, particularly between Sankari and Salem. Planning is essential to ensure communication devices are capable of operating over multiple networks.

## 10. Estimated Emergency Response Times for Different Route Segments

- **Urban (e.g., Salem):** 15-20 minutes due to proximity to emergency services.
- **Rural:** 30-60 minutes, depending on location. Emergency services may take longer to reach remote areas.

## 12. Overall Summary of Risk Assessment

This route presents moderate risks primarily related to traffic congestion and weather-related disruptions. The infrastructure is adequate but needs monitoring for potholes and bottlenecks. Emergency plans should include alternative routes and network reliability checks for communication. Adherence to safety regulations and cautious driving will minimize incident risks, particularly with hazardous materials.

Overall, this is a viable route for transporting hazardous materials, given that all safety protocols are diligently followed.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.63266, 78.12699 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 3  | Turn       | Medium     | 11.44900, 77.87406 | 30 KM/Hr    |
| 4  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 7  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.45571, 77.81702 | 30 KM/Hr    |
| 9  | Turn       | High       | 11.45615, 77.81677 | 15 KM/Hr    |
| 10 | Turn       | Medium     | 11.53552, 77.95763 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.61233, 78.10397 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 12 | Turn      | High       | 11.61216, 78.10423 | 15 KM/Hr    |
| 13 | Turn      | High       | 11.63300, 78.12746 | 15 KM/Hr    |
| 14 | Turn      | Medium     | 11.65825, 78.20469 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.66366, 78.20649 | 30 KM/Hr    |
| 16 | Turn      | Medium     | 11.67129, 78.22978 | 30 KM/Hr    |
| 17 | Turn      | High       | 11.67120, 78.23136 | 15 KM/Hr    |
| 18 | Turn      | High       | 11.67182, 78.23141 | 15 KM/Hr    |
| 19 | Turn      | Medium     | 11.67422, 78.23881 | 30 KM/Hr    |
| 20 | Turn      | Medium     | 11.68017, 78.24250 | 30 KM/Hr    |
| 21 | Turn      | Medium     | 11.83514, 78.35305 | 30 KM/Hr    |
| 22 | Turn      | High       | 11.93905, 78.40417 | 15 KM/Hr    |
| 23 | Turn      | Medium     | 11.93960, 78.40395 | 30 KM/Hr    |
| 24 | Turn      | Medium     | 11.94039, 78.40200 | 30 KM/Hr    |
| 25 | Turn      | Medium     | 11.95570, 78.39975 | 30 KM/Hr    |
| 26 | Turn      | Medium     | 11.98751, 78.39991 | 30 KM/Hr    |
| 27 | Turn      | Medium     | 11.98876, 78.39941 | 30 KM/Hr    |
| 28 | Turn      | High       | 12.01020, 78.39698 | 15 KM/Hr    |
| 29 | Turn      | Medium     | 12.01049, 78.39755 | 30 KM/Hr    |
| 30 | Turn      | Medium     | 12.01222, 78.39878 | 30 KM/Hr    |
| 31 | Turn      | Medium     | 12.01366, 78.39888 | 30 KM/Hr    |
| 32 | Turn      | Medium     | 12.01431, 78.39858 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                                                            | coordinates               | speed_limit | risk_level |
|---|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 2 | hospital | C. N. Hospital                                                  | 11.55724,<br>78.0096      | 30 km/h     | Medium     |
| 3 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 4 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 5 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |

|    | type     | name                                                   | coordinates               | speed_limit | risk_level |
|----|----------|--------------------------------------------------------|---------------------------|-------------|------------|
| 8  | hospital | Vinayaka Mission's Kirupananda Variyar Medical College | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 12 | hospital | Thayammal Hospitals                                    | 11.6237472,<br>78.1177631 | 30 km/h     | Medium     |
| 13 | hospital | Vanitha Hospital                                       | 11.6239931,<br>78.1184622 | 30 km/h     | Medium     |
| 15 | hospital | Priyam Speciality Hospital                             | 11.63446,<br>78.124388    | 30 km/h     | Medium     |
| 16 | hospital | Universal Cancer Hospital                              | 11.6336739,<br>78.1267767 | 30 km/h     | Medium     |
| 17 | hospital | AVM Hospital                                           | 11.6264369,<br>78.1394204 | 30 km/h     | Medium     |
| 18 | hospital | மருத்துவமனை நிலம்                                      | 11.625344,<br>78.146058   | 30 km/h     | Medium     |
| 19 | hospital | Dharan Hospital                                        | 11.6252302,<br>78.1451753 | 30 km/h     | Medium     |
| 20 | hospital | Udhayam Hospital                                       | 11.625819,<br>78.145958   | 30 km/h     | Medium     |
| 21 | pharmacy | மருந்தகம்                                              | 11.6251501,<br>78.1470098 | 30 km/h     | Medium     |
| 22 | hospital | Rps Hospital                                           | 11.6260528,<br>78.1474921 | 30 km/h     | Medium     |
| 23 | hospital | Akitha Hospital                                        | 11.6256027,<br>78.1489875 | 30 km/h     | Medium     |
| 24 | hospital | Deepam Speciality Hospital                             | 11.6269734,<br>78.1476357 | 30 km/h     | Medium     |
| 25 | hospital | Manima Clinic                                          | 11.623248,<br>78.1465721  | 30 km/h     | Medium     |
| 28 | hospital | M. Thathanur Primary Health Center                     | 11.6824017,<br>78.2445813 | 30 km/h     | Medium     |
| 29 | hospital | Goverment Hospital                                     | 11.7580777,<br>78.2969128 | 30 km/h     | Medium     |

**Crowded Spots**

|    | type       | name                                                    | coordinates               | speed_limit | risk_level |
|----|------------|---------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school     | KRP Matric. Hr. Sec School                              | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 6  | school     | Vinayaka Missions Annapoorani Matric School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 7  | university | Vinayaka Missions Vinayaka Missions Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 9  | school     | Dist Education Training                                 | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 10 | school     | DIET - Salem                                            | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 11 | school     | Sri Vidhya Bharati Matric HR Sec School                 | 11.6186069,<br>78.1103239 | 30 km/h     | Medium     |
| 14 | school     | Sowdeshwari College                                     | 11.6330643,<br>78.1276021 | 30 km/h     | Medium     |
| 26 | school     | Ganesh College                                          | 11.6566575,<br>78.1991121 | 30 km/h     | Medium     |
| 27 | school     | MettupattiThathanur Primary School                      | 11.6827707,<br>78.2442509 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.44900, 77.87406





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45571, 77.81702





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45615, 77.81677

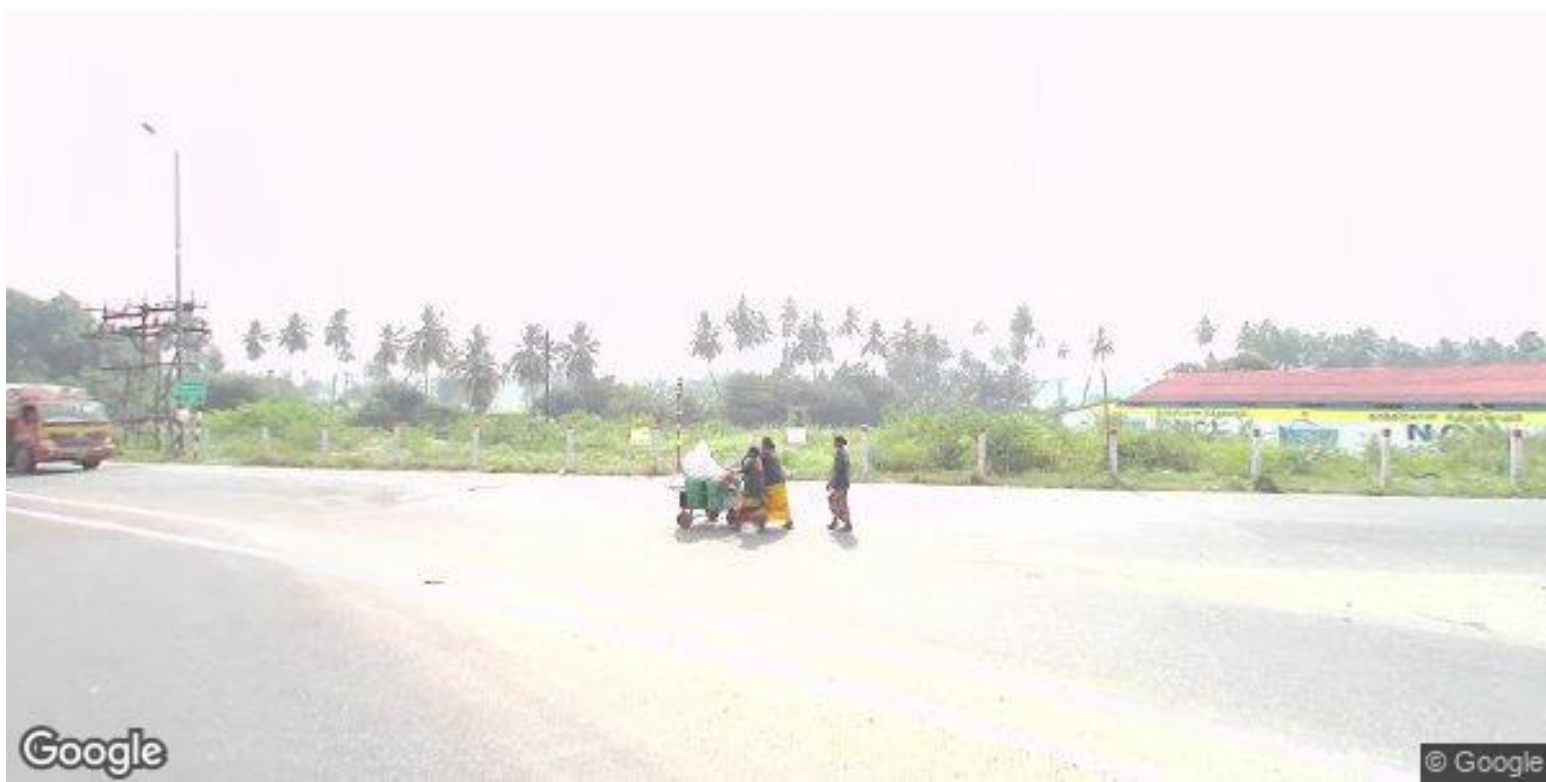


**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53552, 77.95763



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.61233, 78.10397





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.61216, 78.10423



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.63300, 78.12746





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.65825, 78.20469



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**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67120, 78.23136



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67182, 78.23141





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.68017, 78.24250



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**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.83514, 78.35305



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.93905, 78.40417





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.93960, 78.40395



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.94039, 78.40200





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.95570, 78.39975



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.98751, 78.39991



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.98876, 78.39941





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.01020, 78.39698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01049, 78.39755





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01222, 78.39878



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01366, 78.39888



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01431, 78.39858

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